

Tennessee Comprehensive Assessment Program

TCAP

Science
Grade 5
Practice Items



The distances of several stars from Earth are shown in the table.

Star Distances

Star	Distance from Earth (light-years)
Capella	43
Sirius	9
Polaris	434
Spica	261

Which of these describes what will be seen when looking at the stars from Earth?

- A. Capella appears brighter than Sirius but dimmer than Polaris when viewed from Earth.
- B. Spica appears dimmer than Polaris but brighter than Capella when viewed from Earth.
- C. Capella appears brighter than Spica but dimmer than Sirius when viewed from Earth.
- D. Polaris appears brighter than Spica but dimmer than Capella when viewed from Earth.

A toy car moves at a constant rate along a level surface. A student measures the distance the toy car moves in centimeters every ten seconds. The measurements are recorded in the data table.

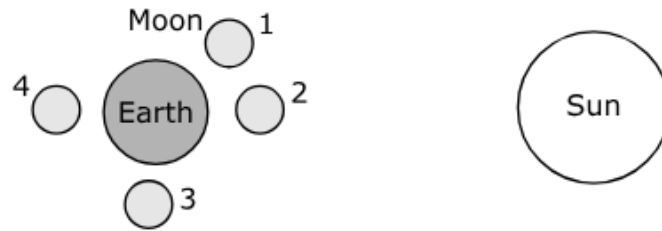
Toy Car Motion

Measurement	1	2	3	4	5	6
Distance (cm)	4	8	12	16	___?	___?
Time (s)	10	20	30	40	50	60

Which **two** distances are the **best** predictions to complete the table?

- M. 18 cm
- P. 20 cm
- R. 24 cm
- S. 28 cm
- T. 30 cm

The diagram shows Earth, the sun, and four positions the moon might be in as it revolves around Earth.

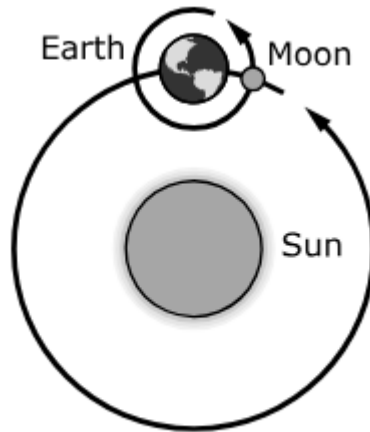


Note: Figure is not drawn to scale.

A solar eclipse is **most likely** to occur when the moon is in which position?

- A. position 1
- B. position 2
- C. position 3
- D. position 4

Gravity is responsible for the motion of Earth, the moon, and the sun.

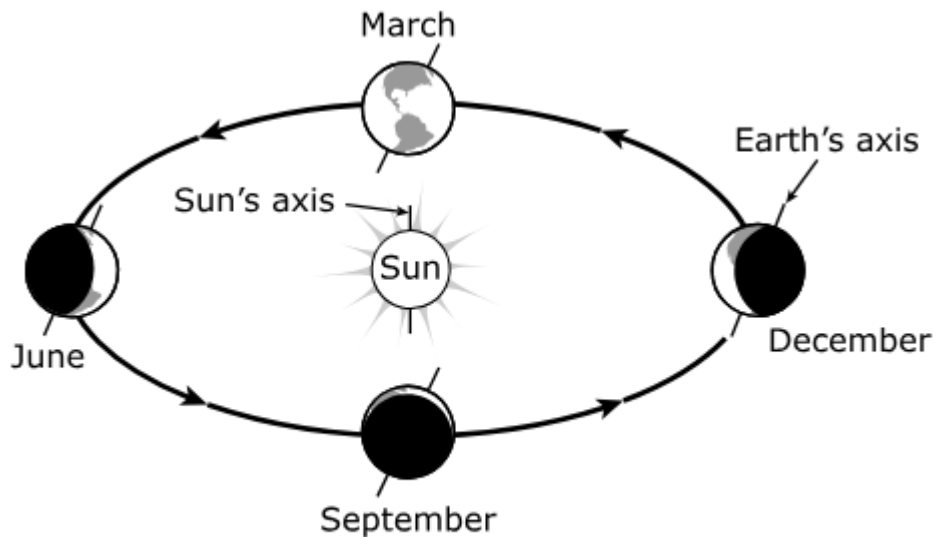


Note: Figure is not drawn to scale.

Which **three** statements correctly describe how gravity is affected by mass and distance?

- M. The gravity between Earth and the moon would be stronger if the moon were closer to Earth.
- P. The gravity between the sun and Earth would be weaker if Earth were closer to the sun.
- R. The gravity between Earth and the moon would be stronger if the moon had more mass.
- S. The gravity on the surface of the moon would be weaker if the moon stayed the same size but had more mass.
- T. The gravity on the surface of Earth would be stronger if the distance to the center of Earth were smaller but had the same mass.

Parts of Earth receive different amounts of daylight during the year. The diagram shows Earth in different positions in its revolution around the sun.

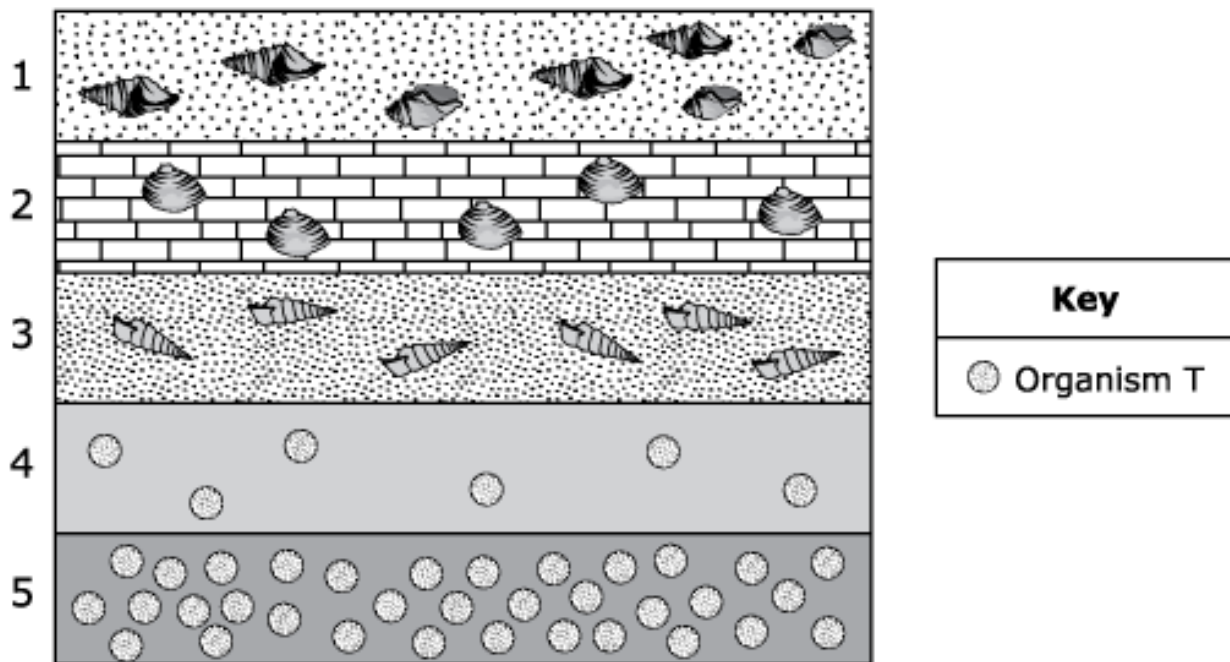


Note: Figure is not drawn to scale.

The changes in the number of daylight hours on Earth are caused by

- A. the tilt of Earth's axis.
- B. the tilt of the sun's axis.
- C. the distance of Earth from the sun.
- D. the movement of the sun around Earth

Organism T became extinct about 250 million years ago. The rock strata show fossil records for Organism T and some other organisms.



Because Organism T was found in rock layers 4 and 5, what conclusion can be made?

- M. Layer 1 is the oldest of the rock strata in the diagram.
- P. Layer 3 was formed less than 250 million years ago.
- R. The formation of layer 5 occurred less than 250 million years ago.
- S. Organism T existed on Earth for only a few years.

Students place four bean seeds into a pot. All of the seeds sprout roots and shoots. The plants have green leaves and stems with white flowers. The mature plants produce long bean pods.

Which **three** plant characteristics are **most likely** a result of interactions with the environment?

- A. All of the seeds sprout roots that grow down.
- B. If the plants are not watered, they start to droop.
- C. The mature plants produce long green pods containing beans.
- D. The plants have green stems, green leaves, and white flowers.
- E. The shoots all bend to grow toward the only source of sunlight.

Students wanted to learn what happens when substances are mixed. They mixed liquids and solids for their experiment. Their observations are recorded in the table.

Students' Data

Mixture	Liquid	Solid	Observations
1	Water (no color)	Sugar (white)	Forms a liquid with no color
2	Vinegar (no color)	Baking soda (white)	Gas bubbles out of a liquid with no color

Which mixture went through changes that produced new substances?

- M. Mixture 1, because it formed only a liquid
- P. Mixture 1, because it had no color
- R. Mixture 2, because bubbles formed
- S. Mixture 2, because no solid formed

The table lists objects found in space.

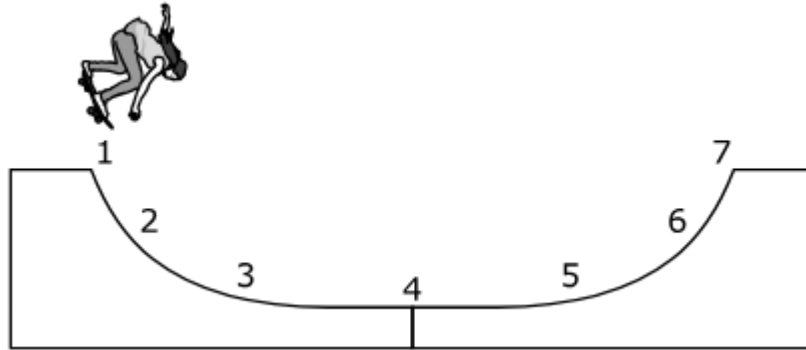
Objects in Space

Object	Location	Description
1	Orbiting the sun between Mars and Jupiter	Large rocky body with many craters
2	Orbiting the sun beyond Neptune	Small ball of ice, rock, and dust with two tails
3	Orbiting Saturn	Large rocky body with an atmosphere
4	Orbiting Earth	Large rocky body with many craters

Which objects are most likely moons?

- A. Objects 1 and 2.
- B. Objects 2 and 4.
- C. Objects 3 and 4.
- D. Objects 1 and 3.

A local skate park has a U-shaped ramp, as shown. A student rides her skateboard back and forth along the ramp at a steady pace. She reaches the next location every 0.4 seconds.



The student starts a timer when she leaves Location 1. How many seconds will it take her to return to Location 1?

- M. 4.4 seconds
- P. 4.6 seconds
- R. 4.8 seconds
- S. 5.6 seconds

Four boats are being loaded at a loading dock. As the boats are getting loaded, the boats are sinking deeper into the water. The table shows the depths that the boats sank into the water.

Boats on the Dock

Boat	Depth Sunk (m)
1	0.53
2	1.05
3	1.32
4	1.56

Which boat experiences the **greatest** force of gravity from Earth?

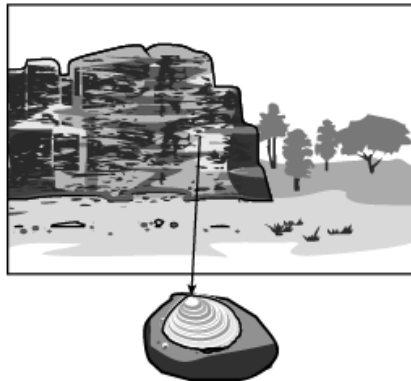
- A. Boat 1
- B. Boat 2
- C. Boat 3
- D. Boat 4

The questions on pages 14 -15 refer to the passage(s) and image(s) shown.

Clam Fossils in Tennessee – Part 1

Students went on a fossil hunting trip in western Tennessee. They saw a fossil in the layers of a rock cliff near a stream. Figure 1 shows the place where they saw this fossil.

Figure 1.
Clam Fossil Location

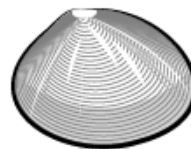


The students learned that the fossil is a clam. This type of clam is no longer found on Earth. However, a similar living species of clam can be found in some places on Earth. Figure 2 shows the clam fossil species and the living clam species.

Figure 2.
Clam Fossil and Living Clam



Clam fossil species
from Tennessee



Living clam
species

The students learn that scientists have compared the clam fossil found in Tennessee to the living clam. Table 1 gives information about the clam fossil species and the living clam species.

Table 1. Information About the Clam Fossil and the Living Clam

	Clam Fossil	Living Clam
Current Environment Where Clam Was Found	- In a rock cliff at the edge of a forest	- On the floor of the ocean
Parts of Body	- Two shells joined together at one end - Other body parts are not preserved	- Two shells joined together at one end - Muscle opens and closes shell - Foot digs into ocean floor to hide clam in sand or mud

Clam Fossils in Tennessee – Part 2

The students also learned that members of a clam species can differ from each other. Table 2 compares two groups that are part of the same species of clam.

Table 2. Comparison of Two Groups of Living Clams

	Living Clam Group 1	Living Clam Group 2
Environment Where Clam Is Found	Warm water	Warm water
Place Where Clam Is Buried	Sandy ocean floor	Muddy ocean floor
Number of Ridges on Shell	Older clams have more than younger clams	Older clams have more than younger clams
Size of Ridges on Shell	Big	Small
Thickness of Shell	Thick	Thin

1. Based on the information in Table 1 (Part 1) and Table 2 (Part 2), which statement describes the **most likely** environment in western Tennessee at the time that the fossilized clam was alive?
 - A. The environment had many trees, because modern day clams live in trees.
 - B. The environment had many grasses, because modern day clams live in fields.
 - C. The environment was covered in rock, because modern day clams live in caves.
 - D. The environment was covered in water, because modern day clams live in water.

2. A student claims that the fossil clam is closely related to the living clam. Using the information in Table 1 (Part 1), which **two** statements support the student's claim?
 - M. The claim is supported because the clam fossil was found in a place that is far from the ocean.
 - P. The claim is supported because the clam fossil has the same shape as the living clam.
 - R. The claim is supported because only the shell of the clam fossil is preserved.
 - S. The claim is supported because the clam fossil has two shells.
 - T. The claim is supported because the fossil clam also had a foot.

3. Which claim about how a physical characteristic would give one type of clams an advantage over another type of clams is **best** supported by the information in Table 2 (Part 2)?
 - A. A clam with a thin shell would have an advantage in warm water, compared to a clam with many ridges on its shell.
 - B. A clam with a thick shell would have an advantage in warm water, compared to a clam with many ridges on its shell.
 - C. A clam with small ridges on its shell would have an advantage in muddy places, compared to a clam with a thin shell.
 - D. A clam with big ridges on its shell would have an advantage in sandy places, compared to a clam with small ridges on its shell.

4. One student finds a second clam fossil and notices that this fossil has a much thicker shell than the first fossil. This student claims that having a thicker shell would allow this clam to better survive over a clam with a thinner shell. Using the information in Table 2 (Part 2) which statement **best** supports this student's claim?

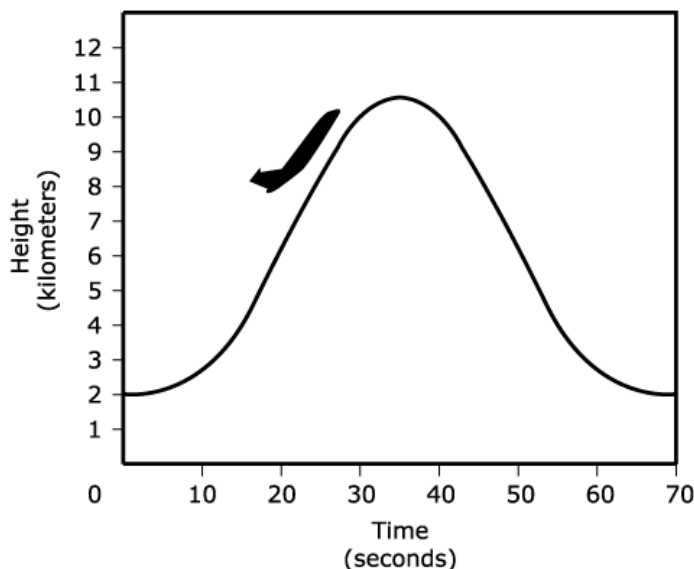
- M. Having a thicker shell would have helped protect the clam from being eaten.
- P. Having a thicker shell would have allowed the clam to dig into the ocean floor.
- R. Having a thicker shell would have allowed the clam to live in warm water.
- S. Having a thicker shell would have helped the clam to get food.

The questions on pages 18 -19 refer to the passage(s) and image(s) shown.

Astronaut Training – Part 1

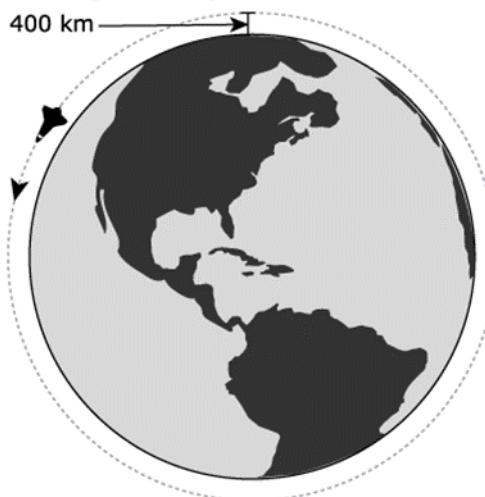
One way that astronauts train for spaceflight is on an airplane that flies up and down in steep curves. Astronauts train for about an hour each day on the airplane while it flies up and down as many as 60 times. Figure 1 shows how the airplane flies. Astronauts start to feel weightless at the point in the flight shown by the picture of the airplane.

Figure 1. Astronaut Training Airplane



Actual spaceflights usually take astronauts into orbit, where they may stay for a few days or even several months. Astronauts in orbit circle Earth as many as 16 times each day. Astronauts feel weightless any time they are in orbit. Figure 2 shows an example of a spacecraft in orbit around Earth.

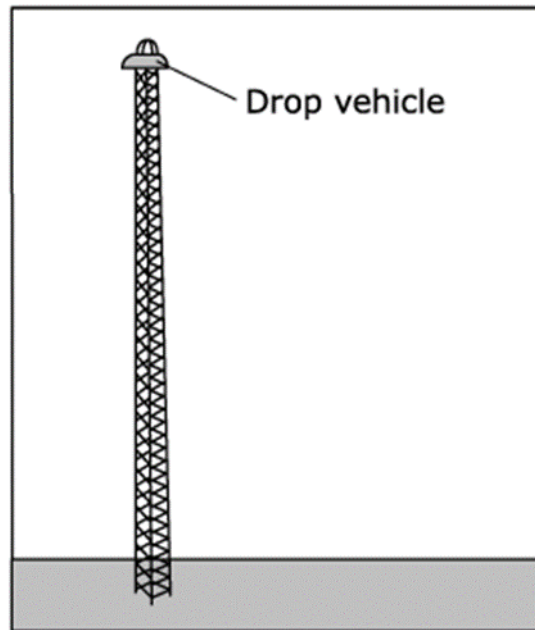
Figure 2. Spacecraft in Orbit



Astronaut Training – Part 2

People who want to feel weightless like an astronaut can ride on a drop tower. The ride takes people to the top of a tall tower. The drop vehicle and the people fall for about three seconds. Near the bottom of the tower, the drop vehicle slows to a stop. Figure 3 shows the drop tower with the drop vehicle at the top of the tower.

Figure 3. Drop Tower



1. Which statement **best** compares the force of gravity acting on the airplane in Figure 1 (Part 1) and the spacecraft in Figure 2 (Part 1)?
- M. the force of gravity on the plane is upwards while the force of gravity on the spacecraft is forwards
 - P. the force of gravity is towards the center of the earth for both the airplane and the spacecraft
 - R. the force of gravity is forwards for the airplane and upwards for the spacecraft
 - S. the force of gravity is towards the surface of the earth for both the airplane and the spacecraft
2. Which claim is **most** supported by evidence in Figure 1 and Figure 2 (Part 1)?
- A. Earth's gravity only affects objects on the surface of Earth.
 - B. An object being pulled toward Earth's center can move in a curved path.
 - C. Earth only pulls on objects when they are moving away from the surface of Earth.
 - D. Earth's gravity is stronger in space.
3. Based on Figure 3 (Part 2), which statement **best** describes why the people riding the drop tower can feel weightless?
- M. The people and the drop vehicle are both moving in the direction that gravity pulls.
 - P. The people and the drop vehicle are both moving too quickly for gravity to affect them.
 - R. The force of gravity decreases because the people and the drop vehicle move in opposite directions.
 - S. The force of gravity between Earth and both the people and the drop vehicle is too small to affect them.

4. When the drop vehicle approaches the bottom of the drop tower, the drop vehicle begins slowing down. During this time, the force on the people is much greater than the force of gravity. Based on Figure 3 (Part 2), which statement **best** describes a reason for this increase in force?
- A. The drop vehicle is at the closest point in its drop to Earth's surface.
 - B. The force slowing the drop vehicle is in a direction that is away from Earth's center.
 - C. The force of gravity on the people increases because the drop vehicle is moving toward Earth's surface.
 - D. The force of gravity starts to pull on the people when the drop vehicle begins moving away from Earth's center.

Metadata – Science Items

Page Number	Item Type	Key	TN Standards
1	MC	C	5.ESS1.1
2	MS	P, R	5.PS2.2
3	MC	B	5.ESS1.4
4	MS	M, R, T	5.PS2.4
5	MC	A	5.ESS1.5
6	MC	P	5.ESS1.7
7	MS	A, B, E	5.LS3.1
8	MC	R	5.PS1.4
9	MC	C	5.ESS1.3
10	MC	R	5.PS2.2
11	MC	D	5.PS2.4

Clam Fossils in Tennessee

Page Number	Item Type	Item Number	Key	TN Standards
12	Stimulus			
13	Stimulus			
14	MC	1	D	5.LS4.1
14	MS	2	P, S	5.LS4.1
14	MC	3	D	5.LS4.2
15	MC	4	M	5.LS4.2

Astronaut Training

Page Number	Item Type	Item Number	Key	TN Standards
16	Stimulus			
17	Stimulus			
18	MC	1	P	5.PS2.3
18	MC	2	B	5.PS2.3
18	MC	3	M	5.PS2.3
19	MC	4	B	5.PS2.3

Metadata Definitions

Item Type	MC = Multiple Choice MS = Multiple Select Stimulus = Cluster support information
Key	Correct Answer
TN Standards	Primary standard assessed in item.